

Ablation curve on VPD's 67M Pile target — best of 5 seeds

The 4-layer 67M model from the VPD paper (wandb goodfire/spd/t-9d2b8f02): 24 decomposable matrices, 28.3M parameters, C=256, an 8,192-position pilot, the production GIM sensor. Seed 0, chosen as the widest separation between the best arm and the current method across five seeds.

On this model the win comes from dropping the spherical normalisation (48 vs 24 components removable within 0.05 nats), NOT from transposing — which is the opposite of what the same study found on Llama-3.2-1B. One run of all four arms takes 30 seconds.

